AKSHAY KUMAR BODDU

Generative AI Engineer | Agentic AI | LLM Applications | RAG & LLM Evaluation
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SUMMARY

Generative AI engineer with ~4 years of software engineering experience, including 2+ years building production LLM agent systems for a **top-10 U.S. commercial mortgage lender**. Designs multi-agent workflows, RAG pipelines, LLM evaluation frameworks, human-in-the-loop review, and MCP-based tool integrations using LangGraph, LangChain, FastAPI, Azure, and AWS. Built document-extraction agents that reach **95–98% accuracy** and cut loan-document review from **~8 hours to ~5 minutes**, plus a solo-built application projected to save **~1,000 hours and ~$117K per year**. Experienced at taking AI features from POC through evaluation, UAT, and production deployment alongside business stakeholders.

TECHNICAL SKILLS

- **AI Platform Engineering:** Agentic AI, Multi-Agent Systems, Retrieval-Augmented Generation (RAG), LLM Evaluation, Human-in-the-Loop Design, Model Context Protocol (MCP), Prompt Orchestration, Structured Output Extraction, Enterprise AI Integration
- **AI Frameworks:** LangChain, LangGraph, CrewAI, FastAPI, LangSmith, Playwright, Streamlit
- **Large Language Models:** OpenAI GPT-4o, Anthropic Claude, Llama, Groq, Ollama
- **Vector Search & Retrieval:** Azure AI Search, FAISS, Chroma, Semantic Chunking, Hybrid Search, Embedding Pipelines
- **AI Engineering Tools:** Cursor, Claude Code, GitHub Copilot, Amazon Q, Kiro
- **Cloud:** AWS S3, Lambda, Glue, Redshift, Azure, ADF, Synapse, Azure App Services, Docker, Kubernetes
- **DevOps:** GitHub Actions, GitLab CI, Azure DevOps, Jenkins, Git
- **Security:** OAuth2, Microsoft Entra ID, JWT
- **Languages:** Python, Java, SQL, C, HTML5/CSS3
- **Frameworks:** FastAPI, React.js
- **Data:** SQLite, MySQL, PostgreSQL, MongoDB, SQL Server, pandas, NumPy, scikit-learn, Google Drive API

EXPERIENCE

Solution Analyst (AI & Enterprise Platform Engineering) — Deloitte *Jul 2024 – Present*
Lake Mary, FL

- Architected and built, as the **sole developer**, the Service Action Request (SER) application for a top-10 U.S. commercial mortgage lender. The LLM agent workflow drafts underwriting briefs from loan-servicer requests (e.g., repair-escrow extensions), replacing a **~150-minute manual process**. The business case projects **~1,000 hours and ~$117K saved per year** on repair-escrow extensions alone (40% efficiency gain for 15–20 underwriters; ~$290K total opportunity).
- Packaging SER as an **MCP server** so the client's production applications can call the underwriting agents as tools across platforms. The first integration is into an existing production application.
- Built a multi-document reconciliation agent that extracts and reconciles environmental (ESA), property condition (PCA), appraisal, and zoning reports (**150–200 pages each**) into one structured output. It cuts per-loan review from **~8 hours to ~5 minutes**.
- Delivered extraction agents for loan verification forms 1005 and 1006, parsing structured sections automatically and routing unstructured sections through **human-in-the-loop** review. Cut 1005 processing from ~15–20 min to **~2 min** and 1006 from 4–8 hrs to **3–5 min**.
- Built LLM evaluation frameworks using business-defined ground truth (**10–50 human-labeled documents per agent**), prompt regression testing, and structured-output validation, reaching **95–98% extraction accuracy** against a 90% target.
- Integrated these agents into the team's **15+ agent LangGraph/LangChain orchestration layer** (4-person team; sole developer of every agent above). Used LangSmith tracing to monitor multi-agent runs, debug agent interactions, and tune prompt orchestration.
- Designed RAG architectures using Azure AI Search, FAISS, Chroma, semantic chunking, hybrid search, and embedding pipelines for accurate retrieval across large enterprise document repositories.
- Integrated and benchmarked multiple LLM providers (OpenAI GPT-4o, Anthropic Claude, Llama, Groq), choosing a model for each task based on accuracy, reasoning quality, latency, and cost.
- Built modular FastAPI services deployed in Docker, secured with Microsoft Entra ID, OAuth2, and JWT, and integrated with the client's internal APIs and business applications.
- Designed cloud-native data pipelines across AWS (S3, Lambda, Glue, Redshift) and Azure (ADF, Synapse) to ingest, transform, and process enterprise datasets feeding AI applications.
- Automated build, test, and deployment of AI services with GitHub Actions, GitLab CI, Azure DevOps, and Jenkins.
- Sole developer of an **AI-adoption dashboard requested by the client's VP** that tracks GitHub Copilot and Amazon Q usage across **5 business units**. Leaders can upload usage reports, compare units, and flag inactive users. Includes an **LLM chatbot** so leaders can query usage data in natural language instead of reading charts.
- Spent ~50% of time in discovery with product managers and business teams, defining which underwriting decisions AI automates and which stay human-reviewed. Led live demos and UAT with underwriters and incorporated their feedback before production deployment.
- Produced architecture documentation, technical design artifacts, proofs of concept, and production-readiness recommendations for the client's enterprise AI adoption.

Assistant System Engineer — TCS (Tata Consultancy Services) *May 2021 – Aug 2022*
Bangalore, India

- Developed enterprise financial platform capabilities using React.js, JavaScript, and modern frontend engineering practices while collaborating with architects and cross-functional Agile teams.
- Delivered scalable application features following Agile development methodologies, contributing to improved system performance, software quality, and reduced post-release defects.

- Participated in enterprise software design discussions, API integrations, testing, and CI/CD practices supporting large-scale banking applications.

Software Developer — Gramtarang Inclusive Development *Aug 2020 – May 2021*
Visakhapatnam, India

- Developed healthcare CRM modules supporting patient registration, electronic health records, and regulated healthcare data workflows for enterprise healthcare applications.
- Collaborated with stakeholders to deliver scalable web-based solutions enabling secure patient data management and streamlined healthcare operations.

CERTIFICATIONS

Microsoft Certified:

  - AB-100: Agentic AI Business Solutions Architect
  - PL-400: Power Platform Developer Associate
  - Azure AI Engineer Associate
- AWS Certified Cloud Practitioner

PERSONAL PROJECTS

Personal Job Application Automation Platform — *In Development*

- Building a self-hosted AI-powered job application automation platform (inspired by Tsenta) to replace manually submitting ~20 applications a day, using the Claude API to generate and tailor applications.
- Designed as a local-only web app with direct ATS integrations (Lever, Workday) for reliable submissions and data privacy.

LinkedIn Job Application Automation Agent — *2026*

- Built an end-to-end automation agent that discovers relevant job postings, extracts requirements, and streamlines application workflows using Playwright browser automation and Llama 3.3 via the Groq API.
- Designed an LLM pipeline to parse job descriptions into structured fields stored in SQLite, with Google Drive API integration for resume management and a Streamlit dashboard for tracking application status.

EDUCATION

M.S. Computer and Information Sciences — Florida International University, Miami, FL *Apr 2024*
B.Tech Computer Science Engineering — Centurion University, Visakhapatnam, India *May 2021*